

Final Exam: Choose ONE of Three Build Projects

One-week, individual, AI-assisted application build. You pick a single option below. All three share the same rules, rubric, and submission format.

FORMAT	RELEASED	DUE	WEIGHT	INSTRUCTOR
1-week take-home build (individual). No written exam.	May 30, 2026 — after final class.	Friday, June 5, 2026 · 11:59 PM PT · Canvas	100 points · 28% of final grade.	Dr. Rahul Makhijani

Rules that apply to every option

- **Build & host:** working, runnable code **deployed to a live public URL**.
- **Live 1:1 demo:** a 10-minute one-on-one walkthrough of your hosted app on Saturday, June 6.
- **6 core capabilities** — missing any one **caps your score at 60/100**.
- Integrate **≥2 BAX-423 techniques** from different lectures and **benchmark** their impact.
- Test against **all 4 provided personas** (plus hidden personas at grading); include a pass/fail table in the brief.
- **AI tools encouraged** (Cursor, ChatGPT, Claude, Copilot). Submit key prompts in prompts.md.
- **Code that does not run scores 0** on rubric dimensions 1–4. Always include a README.
- **Submit a ZIP + live URL:** LastName_FirstName_BAX423_Final.zip — code/, data/ (dataset + offline snapshot; see each option for the minimum), brief.pdf (≤4 pp), prompts.md — **plus a live hosted app link**.
- **Rubric (100 pts):** five dimensions — Data Pipeline (15); Matching & Ranking Quality (20); Adaptive Learning & Course Techniques (15); Hosting & Deployment (20); Technical Brief & Live 1:1 Demo (30).

The three options — pick exactly one

Option A — NutriAI	Option B — JobPilot	Option C — EngageIQ
<i>Automated Diet Plan Builder</i>	<i>Smart Job Matcher & Resume Builder</i>	<i>Smart Engagement Opportunity Scorer</i>
WHAT YOU BUILD A diet-planning app that generates a personalized 7-day meal plan in under 60s, tailored to clinical conditions, allergens, diet, and nutrition targets. PIPELINE Clinical filtering → allergy exclusion → diet preference → candidate generation → ranking & diversity → meal assignment → macro/micro analysis → feedback. DATA SOURCE USDA FoodData Central (Branded ~400k, Foundation, SR Legacy). Preprocess to a clean food database; include a ≥5,000-item offline snapshot with full nutrient profiles. TEST PERSONAS IBS+vegetarian; GERD+gluten-free; T2 diabetes+vegan; hypertension+pescatarian. SIGNATURE DELIVERABLE “Why excluded” explanations + PDF/CSV export of the plan; sub-60s generation.	WHAT YOU BUILD A job matcher & resume builder that ingests postings, matches them to a profile, ranks the best fits, and generates a resume tailored to any selected role. PIPELINE Ingestion → deduplication → embedding-based retrieval → multi-stage ranking (filters + scorer + re-rank) → adaptive learning → tailored resume generation. DATA SOURCE Kaggle postings dataset plus Adzuna or JSearch (RapidAPI) job APIs. Assemble a 20,000–50,000-posting sample with an offline snapshot so graders can run without API access. TEST PERSONAS Career pivoter; new grad; experienced/niche constraints; international/visa-constrained. SIGNATURE DELIVERABLE “Generate Resume” action per job + “why ranked here” explanation; batch market analytics.	WHAT YOU BUILD An engagement opportunity scorer that discovers, scores, and ranks where a professional should invest time online across multiple content platforms. PIPELINE Multi-source ingestion & streaming → dedup → embeddings → engagement scoring & ranking → adaptive learning → trend analytics → exportable brief. DATA SOURCE ≥2 of: GitHub API, GH Archive, Reddit (PRAW), Hacker News API. Build a ≥10,000-record dataset spanning all 15 technical domains, with an offline snapshot. TEST PERSONAS Portfolio builder; DevOps niche community; data journalist/trend spotter; startup founder. SIGNATURE DELIVERABLE Dashboard with ranked opportunities, “Why this?” explanations, and a downloadable engagement brief.